

A Location-Based Boarding Room Recommendation System for University Students

1. Student Features (The Seekers)

Authentication & Profile

- **Sign Up / Login:** Via Email/Password or Google Sign-In.
- **University Verification:** Upload a student ID or use a .edu/.ac.uk email to get a "Verified Student" badge.
- **Profile & Preferences:** Set their budget, preferred distance from campus, and room preferences (e.g., quiet, single room, private bathroom) to feed the recommendation engine.

Search, Map & Discovery

- **Smart Feed (Home):** View top-rated rooms, "Recommended for You" (based on preferences), and "Recently Added".
- **Map View:** See all available rooms as pins on a Google Map around their university.
- **Advanced Filters:** Filter by price range, distance (e.g., < 2km), rating, room type (single/shared), and specific facilities (WiFi, AC, kitchen).
- **Search Bar:** Search by specific location names or street names.

Room Details & Navigation

- **Listing View:** Swipe through high-quality images, read descriptions, see price breakdowns, and view the owner's rules.
- **Reviews & Ratings:** Read feedback and see 1-to-5 star ratings left by previous student tenants.

Interaction & Booking

- **Favorites / Wishlist:** Save rooms they like to compare them later.
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2. Room Owner Features (The Hosts)

Authentication & Verification

- **Sign Up / Login:** Secure login process.
- **Identity Verification:** Upload a National ID or utility bill to get a "Verified Owner" badge (builds trust with students).

Listing Management

- **Add New Room:** * Upload multiple photos directly from their phone.
 - Fill in details (price, deposit amount, description).
 - Select facilities from a checklist.
 - **Drop a Pin:** Use an interactive map to set the exact `GeoPoint` location for the navigation feature.
- **Tenant Preferences:** Specify who they want to rent to (e.g., "Boys Only", "Undergraduates", "No Pets").

- **Status Toggle:** Easily flip a switch to mark a room as "Available" or "Occupied" so it disappears from search results when full.
- **Edit/Delete:** Update price, add new photos, or remove the listing entirely.

Tracking

- **Basic Analytics:** View how many students have looked at their listing or favorited it over the past week.
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3 Admin Features (The Platform Managers)

Dashboard & Analytics

- **Overview:** See total active users (students vs. owners), total live listings, and app usage statistics.

User Management

- **Approve Verifications:** Review uploaded Student IDs and Owner utility bills and click "Approve" or "Reject" to grant verified badges.
- **Ban/Suspend Users:** Block users who violate terms of service, spam, or post fake listings.

Content Moderation

- **Listing Approval:** (Optional) Hold new listings in a "Pending" state until an admin quickly reviews the photos and description to ensure it's appropriate before making it public.
- **Manage Reports:** View a queue of listings or messages flagged by students. Investigate and take down listings if necessary.

System Configuration

- **Manage Universities:** Add new university campuses to the system so students can select them during sign-up.

That is a solid feature list! In Sri Lanka, the "boarding" (Bodi) culture is huge, but it is still very old-school. Most students find rooms by walking around university backstreets looking for hand-written "Room for Rent" signs or by scrolling through endless, unorganized Facebook groups.

Here are the key gaps between your system and the current market leaders in Sri Lanka (like Ikman.lk, Boarding.lk, and CampusBodima.lk).

1. The "Trust & Safety" Gap (Verification)

Existing Systems: On sites like Ikman, anyone can post a room. This leads to "ghost listings" or rooms that look nothing like the photos. Verification is usually just a phone number.

Your Gap: Your University Verification (.edu email/ID) and Owner Identity Verification (Utility bills) are game-changers. In Sri Lanka, students (especially female students and their parents) are extremely concerned about safety. Having a "Verified" badge for both sides builds a level of trust that current platforms lack.

2. The "Discovery" Gap (Smart vs. Static)

Existing Systems: Current apps are just "digital newspapers." You search, you see a list, and you scroll. There is no intelligence behind it.

Your Gap: Your Recommendation Engine (Smart Feed) is a major upgrade. In Sri Lanka, students often don't know the "best" areas near a new campus. An app that says, "Based on your Rs. 15,000 budget and your faculty, here are the top 3 quiet rooms," is much more helpful than a random list of 500 rooms.

3. The "Visual & Map" Gap

Existing Systems: Many local listings have poor photos or no location pins. Students often have to call the owner just to find out where the house actually is.

Your Gap: Your Map View with university-specific pins and High-quality Listing Views solve the "Location Mystery." Sri Lankan students care about walking distance (to save on bus/tuk-tuk fare), so seeing the room's exact distance from their specific university gate is a massive advantage.

4. The "Owner Management" Gap

Existing Systems: Most room owners in Sri Lanka are older individuals who aren't tech-savvy. They post once and forget about it. They don't know how many people are looking at their room.

Your Gap: Basic Analytics for owners is unique. If an owner sees that 100 students viewed their room but 0 favorited it, they might realize their price is too high. This "data-driven" approach doesn't really exist in the local boarding market yet.

5. The "Student-Specific" Filters

Existing Systems: Filters are usually just "Price" and "Location."

Your Gap: Your specific filters for "Quiet environment," "Private bathroom," and "Tenant Preferences" (e.g., Boys Only/Undergraduates) are exactly what Sri Lankan "Bodi" culture looks for. Owners are often very specific about who they rent to (e.g., "Only medical students"), and your app makes that filterable.

Feature	Your System	lkman.lk	ezboarding.lk	FB Groups
Student ID Verification (.edu)	✓	✗	✗	✗
Owner Identity Check (NIC/Bills)	✓	✗	✓	✗
Smart Recommendation Feed	✓	✗	✗	✗
Map View (University Pins)	✓	✗	✓	✗
Real-time "Available" Toggle	✓	✗	✗	✗
Analytics for Room Owners	✓	✗	✗	✗
Specific "Student-Only" Filters	✓	✗	✓	✗
Trust Badges (Verified Users)	✓	✗	✓	✗

Why these "Gaps" matter for you:

- **The Trust Gap:** On Ikman or Facebook, anyone can post anything. By requiring a Student ID or Utility Bill, you solve the #1 fear in Sri Lanka: Scams and Safety.
- **The "Map" Gap:** Most local sites just list the city (e.g., "Malabe"). Students want to see exactly how many meters they have to walk to the lecture hall. Your Interactive Map is a huge win here.
- **The "Old Data" Gap:** On Facebook, you see posts from 3 months ago and don't know if the room is taken. Your Status Toggle (Available/Occupied) ensures students don't waste time calling about full rooms.
- **The "Owner" Gap:** Owners currently have no idea why their room isn't renting. Your Basic Analytics helps them understand if their price is too high or if their photos are bad.